

Introduction to Computer Vision

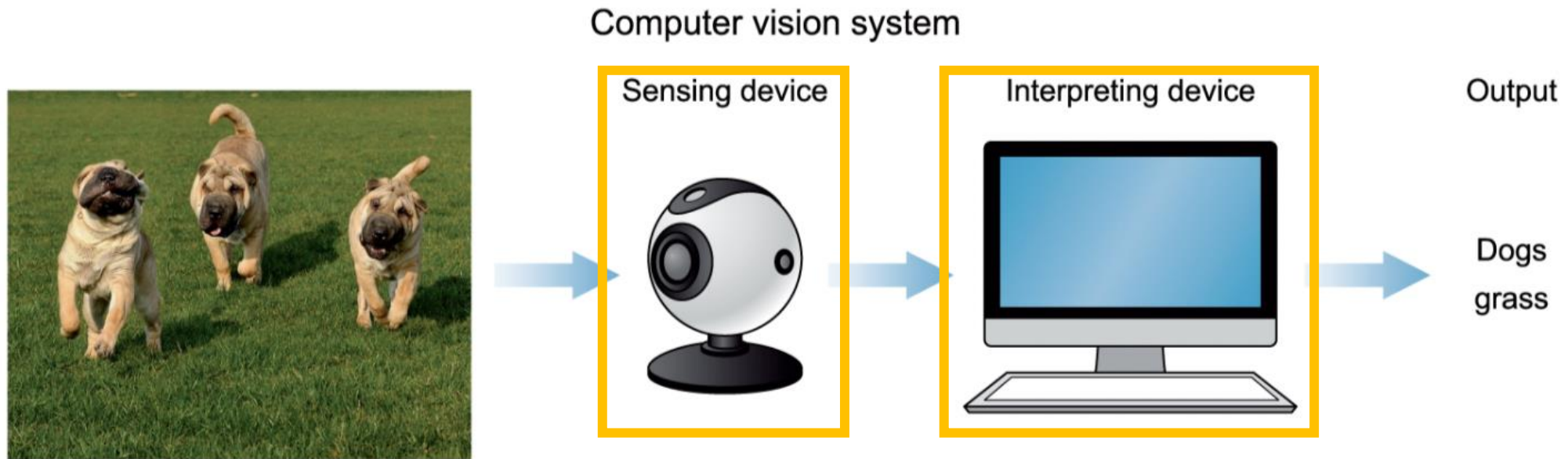
Priyanka Bagade, IITK

Lecture 2

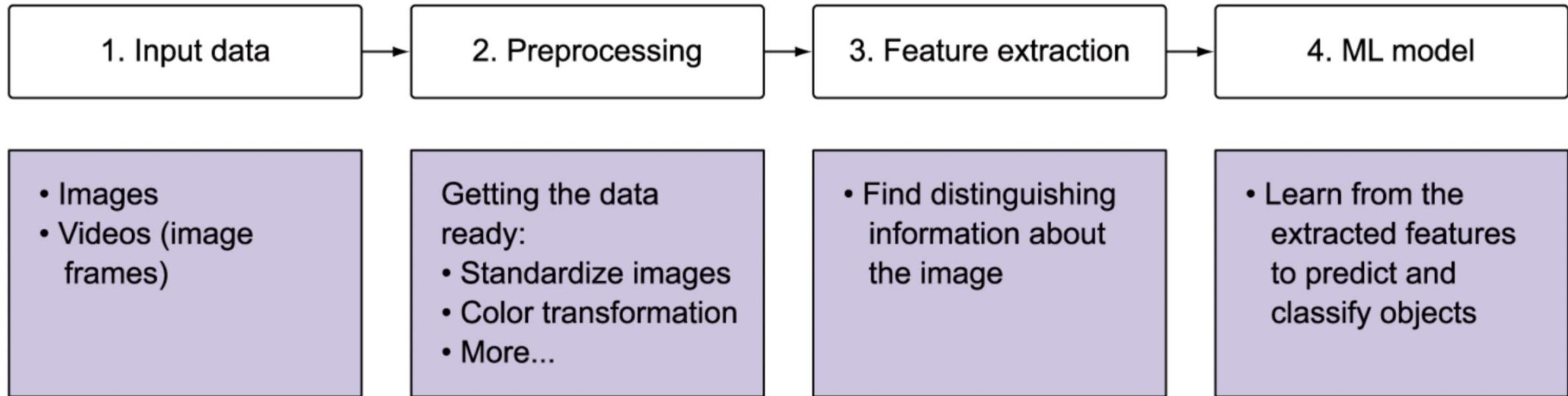
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Computer Vision System

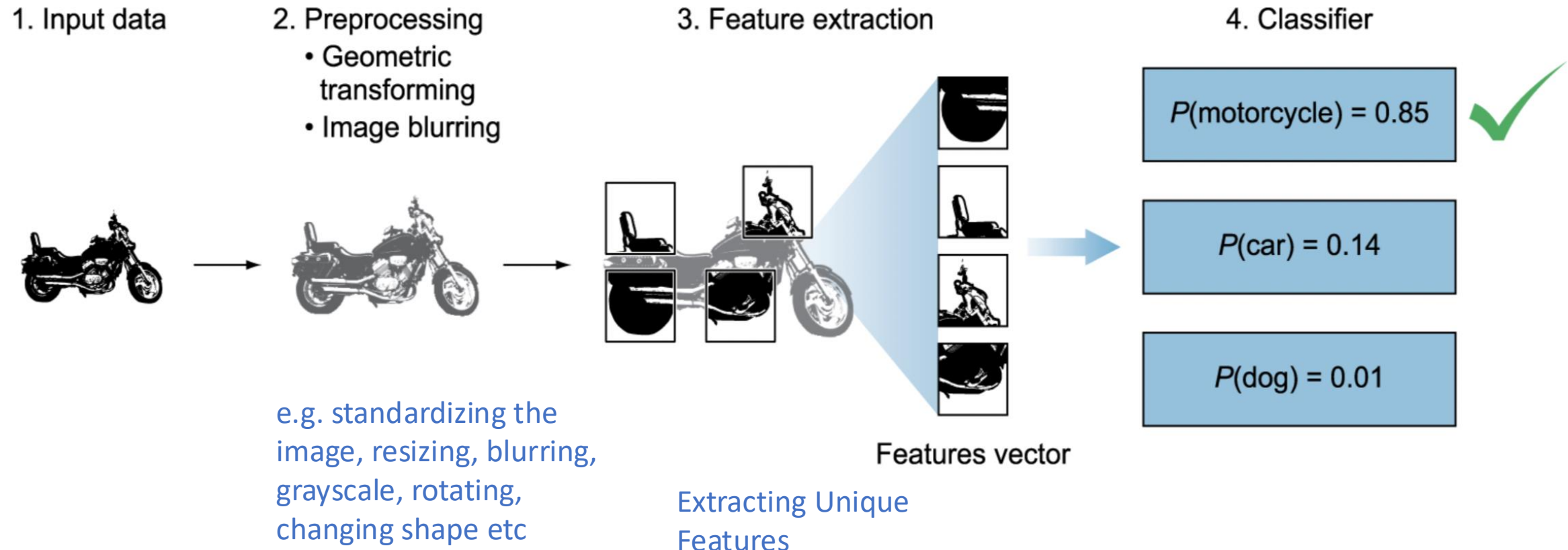


Computer Vision (CV) Pipeline



CV Pipeline using ML Model Example

Classifies the image and outputs the label or class (category)



Improve the Classifier Output

More training data

More preprocessing to reduce noise

Extract better features

Change the classifier algorithm

Tune hyperparameters

Input Image

- Image representation using function of 2 variables: $F(x, y)$
- Each pixel represents intensity of the light
 - Range 0 to 255
 - White = 255
 - Black = 0
 - Grayscale = 1 to 254
 - $F(x, y)$ -> light intensity at position x, y
- Color Image in RGB
 - $F(x, y) = [\text{red}(x, y), \text{green}(x, y), \text{blue}(x, y)]$

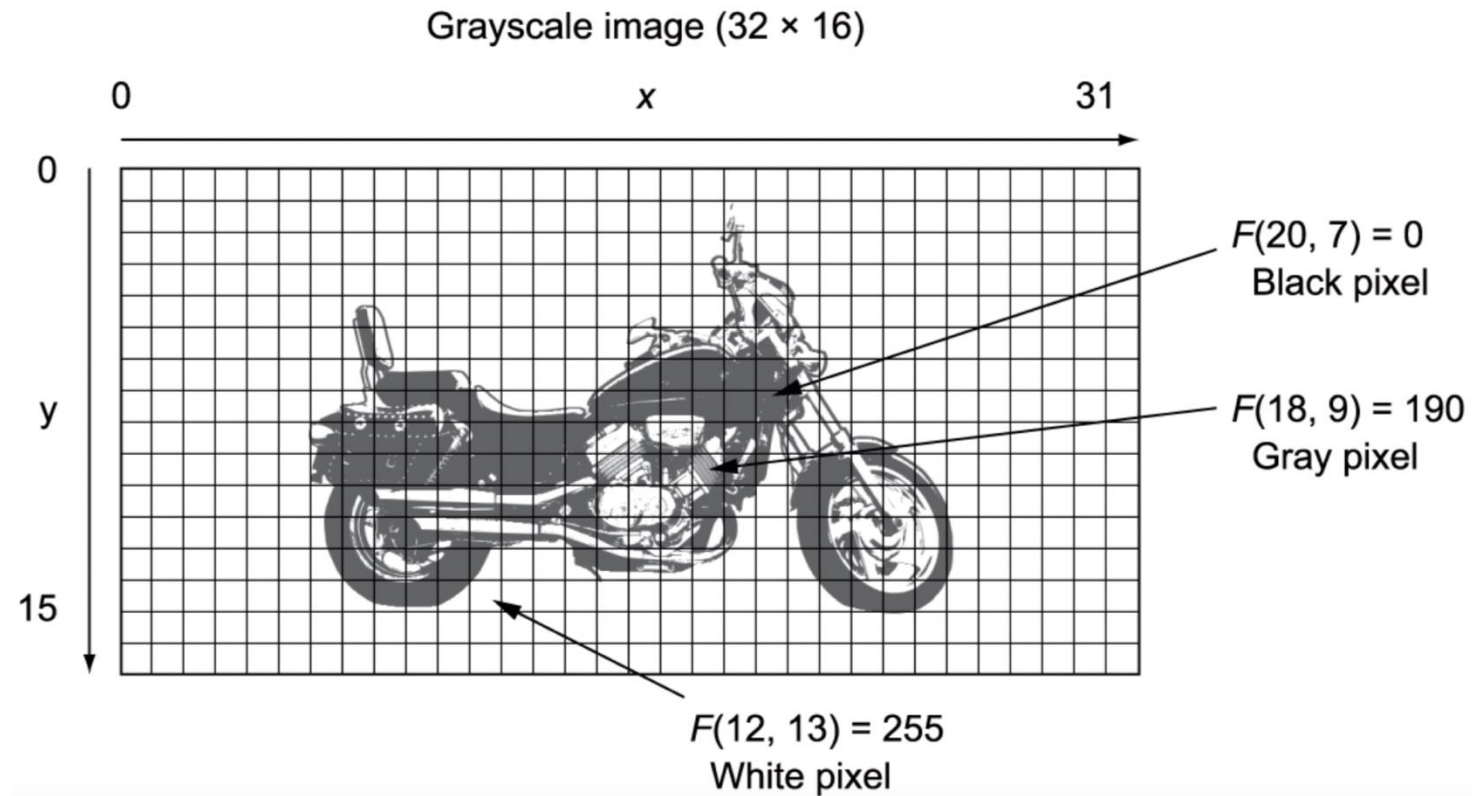


Image Transformations

- Image as a function of $F(x,y)$
- Mathematical operations to transform the input image
 - $F(x,y) \rightarrow G(x,y)$

Application	Transformation
Darken the image.	$G(x, y) = 0.5 * F(x, y)$
Brighten the image.	$G(x, y) = 2 * F(x, y)$
Move an object down 150 pixels.	$G(x, y) = F(x, y - 150)$
Remove the gray in an image to transform the image into black and white.	$G(x, y) = \{ 0 \text{ if } F(x, y) < 130, 255 \text{ otherwise } \}$

How Computers See Images

What we see



What computers see

08	02	22	97	38	15	00	40	00	75	04	05	07	78	52	12	50	77	91	08
49	49	99	40	17	81	18	57	60	87	17	40	98	43	69	46	04	56	62	00
81	49	31	73	55	79	14	29	93	71	40	67	53	99	30	03	49	13	36	65
52	90	95	23	04	60	11	42	69	24	48	56	01	32	54	71	37	02	34	91
22	31	14	71	51	67	43	59	41	92	34	54	22	40	40	28	44	33	13	80
24	47	32	60	99	03	45	02	44	75	33	53	78	36	64	20	35	09	12	80
32	98	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
47	24	20	68	02	62	12	20	95	63	94	39	63	04	49	91	44	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	94	83	14	88	34	89	63	72
21	36	23	09	75	00	74	44	20	45	35	14	00	41	33	97	34	31	33	95
78	17	53	28	22	75	31	67	15	94	03	80	04	42	16	14	09	53	56	92
16	39	05	42	96	35	31	47	55	58	88	24	00	17	54	24	34	29	85	57
84	56	00	48	35	71	89	07	05	44	44	37	44	40	21	58	51	54	17	58
19	80	61	68	05	94	47	49	28	73	92	13	86	52	17	77	04	09	55	40
04	52	08	83	97	35	99	14	07	97	57	32	16	26	26	79	33	27	98	44
04	36	68	81	57	62	20	72	03	16	33	67	46	55	12	32	43	93	53	69
04	42	14	73	38	25	39	11	24	94	72	18	06	46	29	32	40	62	74	36
20	49	34	41	72	30	23	88	34	62	99	69	82	47	59	85	74	04	34	24
20	23	35	29	78	31	90	01	74	31	49	71	48	86	81	14	23	57	05	54
01	70	54	71	83	51	54	49	16	92	33	48	61	43	51	01	89	19	67	48

Color Image Representation

Color image

$F(0, 0) = [11, 102, 35]$



RGB channels

Channel 3
Blue intensity
values

35	165	163	165	158	...
166	166	164	166	159	...
156	158	162	165	159	...
...	...	...	...	...	...

Channel 2
Green intensity
values

102	169	167	169	169	...
170	170	168	170	170	...
160	162	166	169	170	...
...	...	...	...	...	...

Channel 1
Red intensity
values

11	158	156	158	158	...
159	159	157	159	159	...
149	151	155	158	159	...
146	146	149	153	158	...
145	143	143	148	158	...
...	...	...	...	...	...

Green Color Shades

Forest

HEX #0B6623
RGB 11 102 35

Forest green

Codes:
HEX #0B6623
RGB 11 102 35

Mint

HEX #98FB98
RGB 152 251 152

Mint green

Codes:
HEX #98FB98
RGB 152 251 152

Olive

HEX #708238
RGB 112 130 56

Olive green

Codes:
HEX #708238
RGB 112 130 56

Lime

HEX #C7EA46
RGB 199 234 70

Lime green

Codes:
HEX #C7EA46
RGB 199 234 70

Jungle

HEX #29AB87
RGB 41 171 135

Jungle green

Codes:
HEX #29AB87
RGB 41 171 135

Jade

HEX #00A86B
RGB 0 168 107

Jade green

Codes:
HEX #00A86B
RGB 0 168 107

Image Preprocessing

- Data preparation for ML model to analyze and process the data
- Techniques
 - Resizing - CNNs require all input images to be of the same size
 - Change in colors
 - Geometric transformation
- Data standardization and cleaning
 - Data comes from various sources
 - Can be noisy
 - Can't write different algorithms for data coming from different sources
 - Reduces complexity
 - Increases accuracy
- Data Augmentation

Converting color image to Grayscale

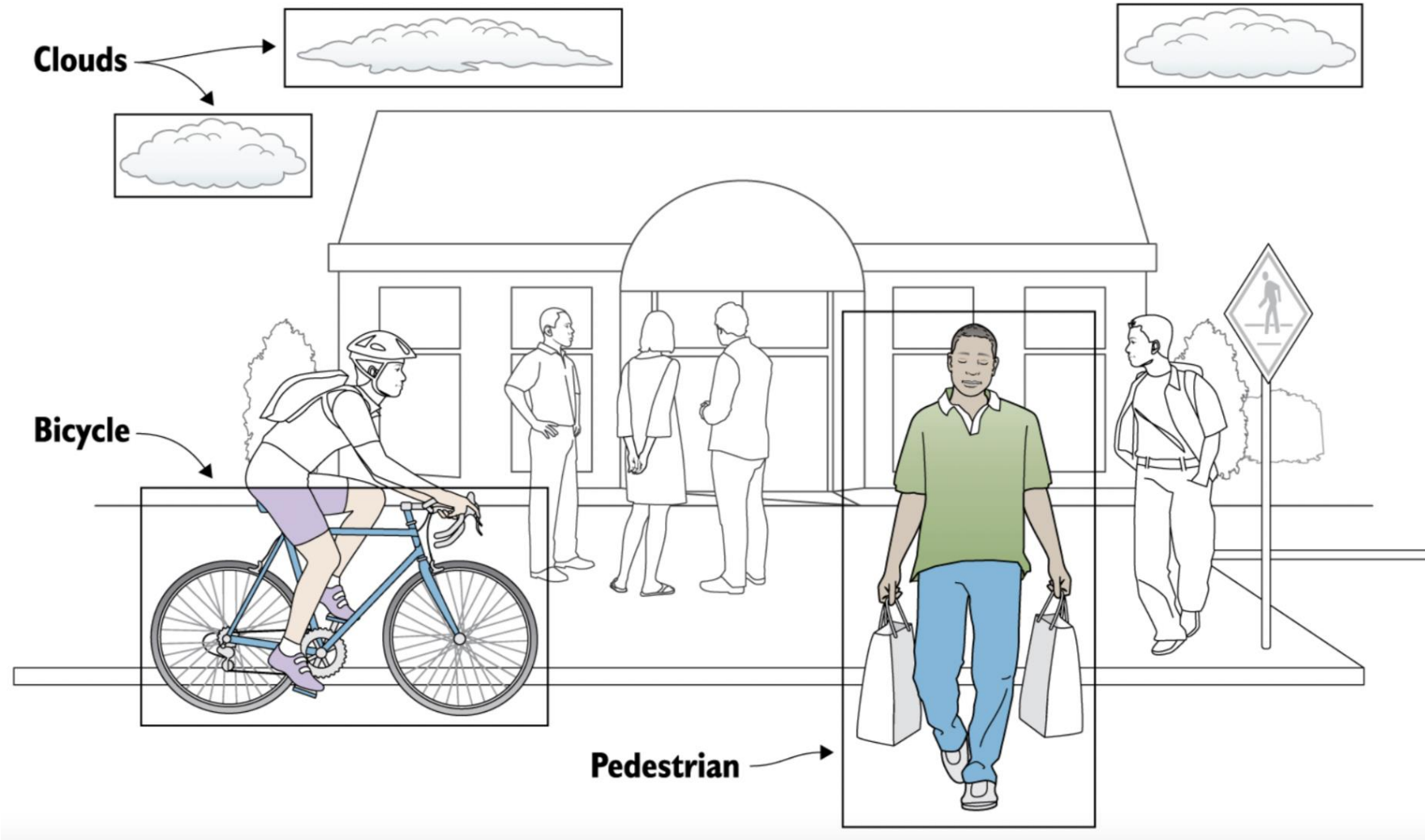
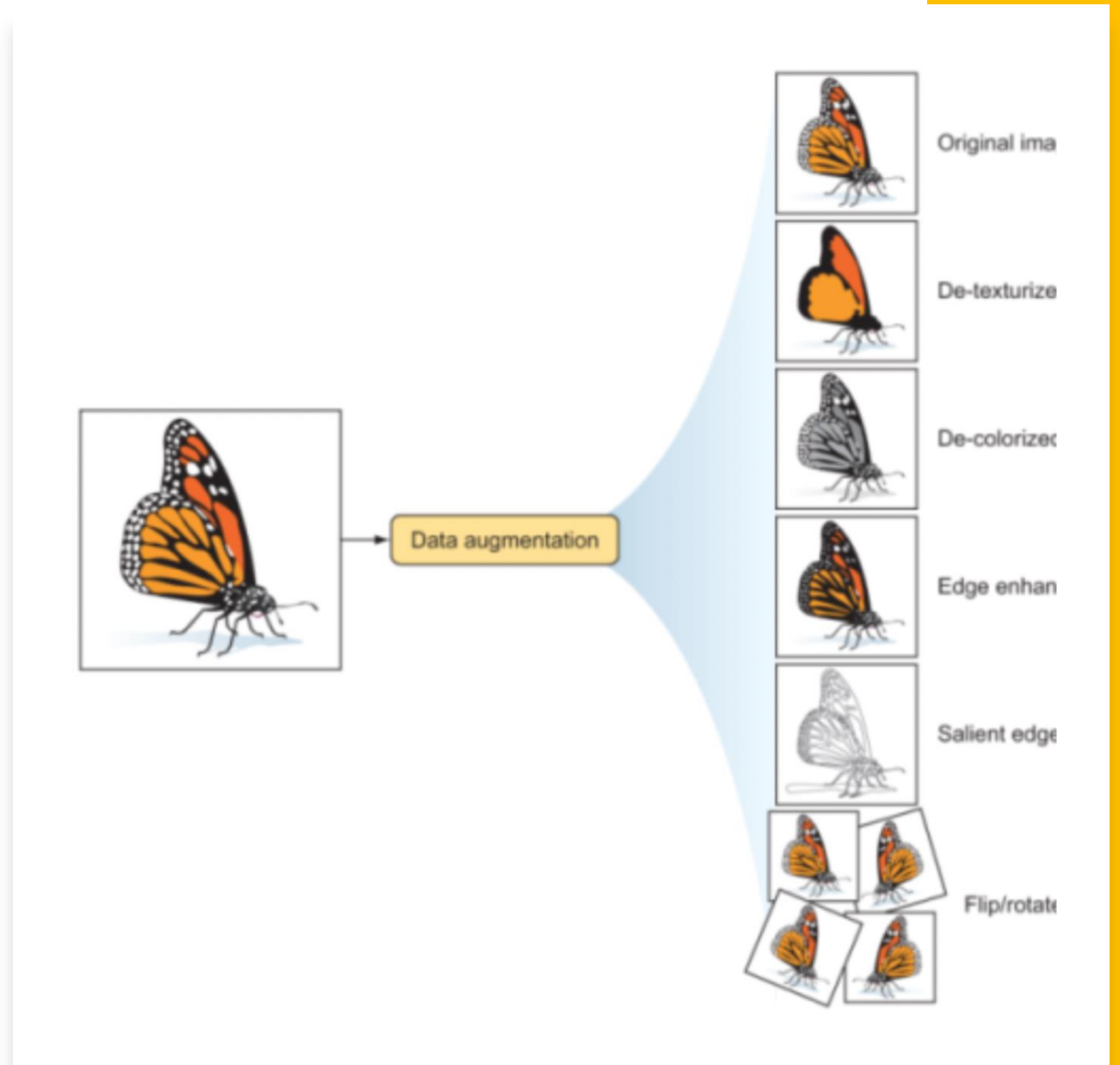


Image preprocessing – Importance of Color images



Image Augmentation

- a technique in machine learning that artificially expands a dataset by creating modified copies of existing images, using transformations like flipping, rotating, zooming, cropping, and color adjustments



Feature Extraction

- Core component of the CV pipeline
- Feature: A feature in computer vision is measurable piece of data in the image that is unique to that specific object
 - E.g. line, edge, image segment, color etc

Feature Extraction – Example

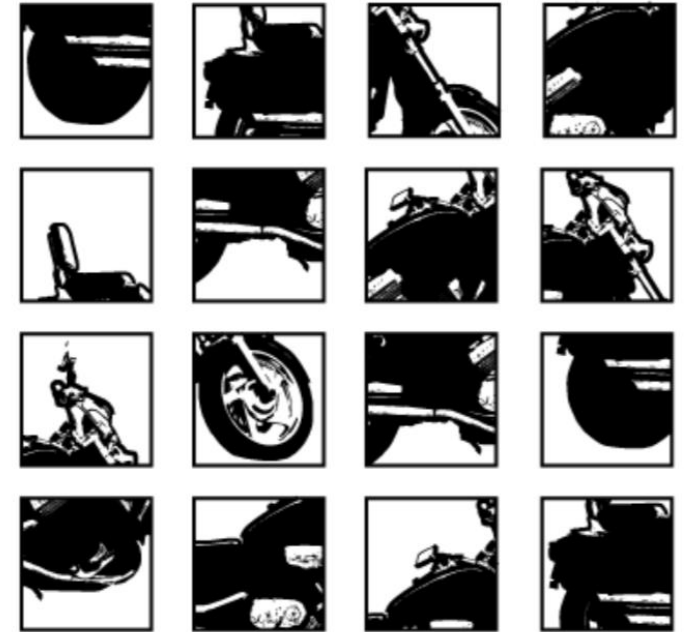
Input data



Feature extraction
algorithm



Features



Feature Generalizability



What makes a good feature?

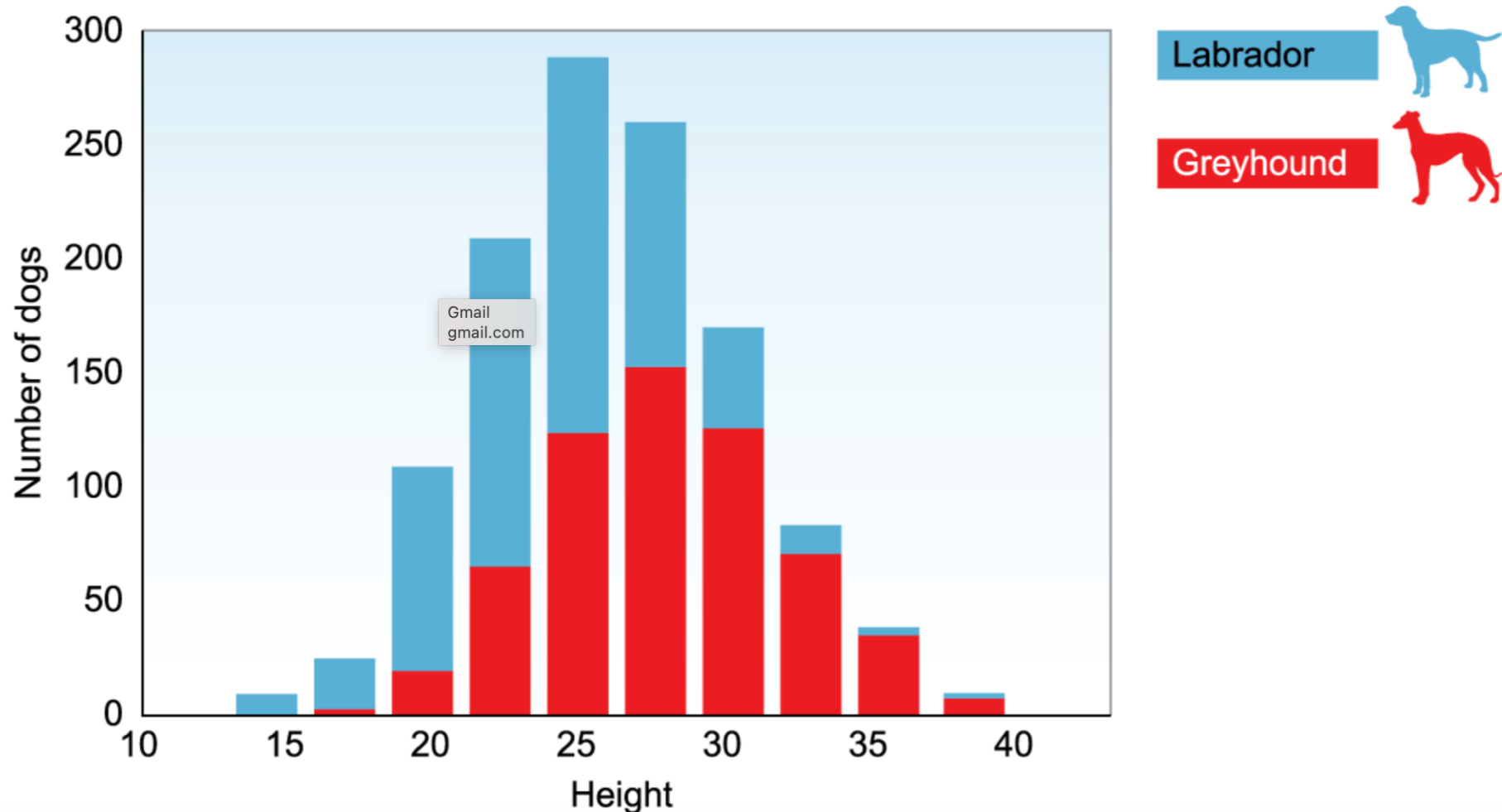
Greyhound



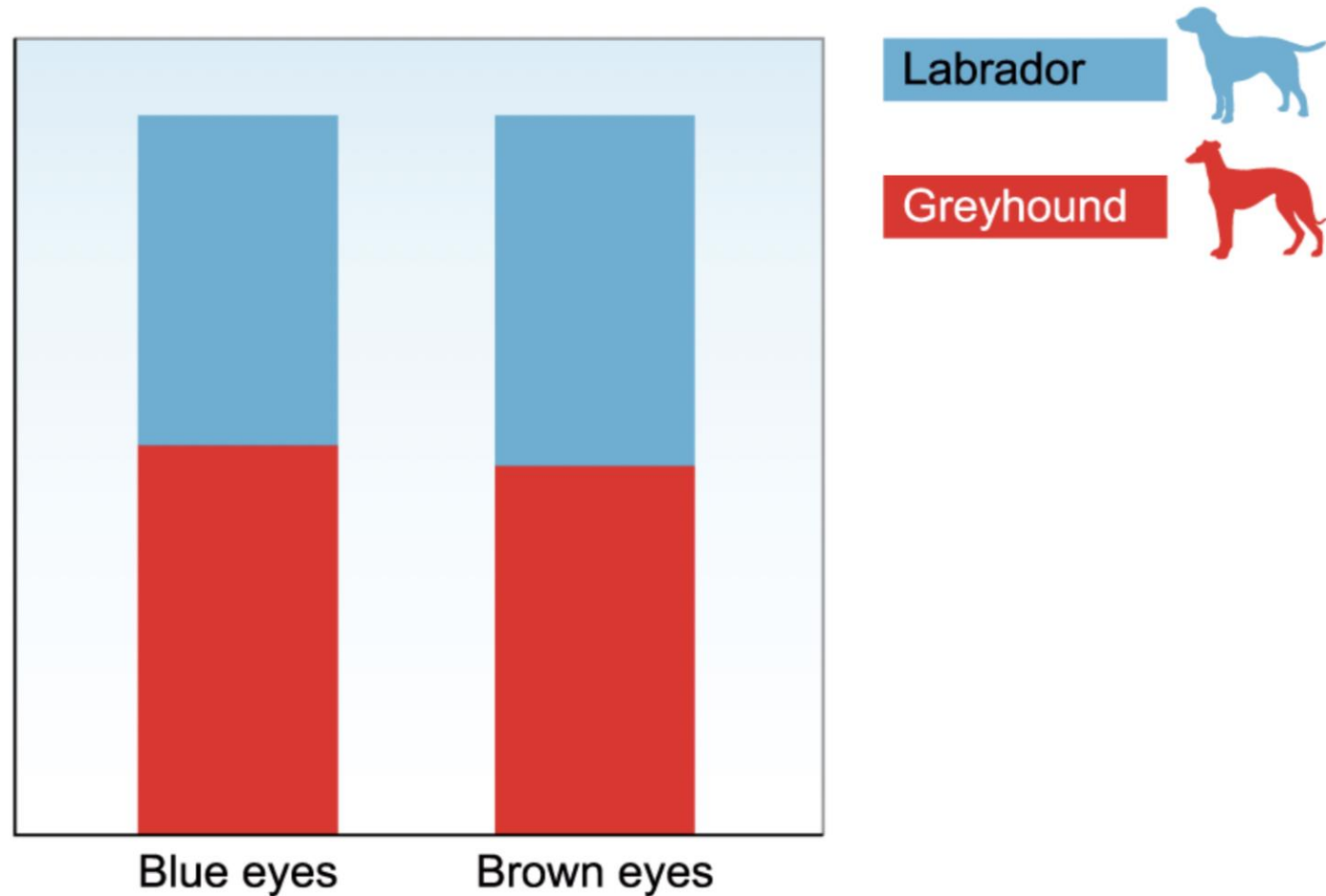
Labrador



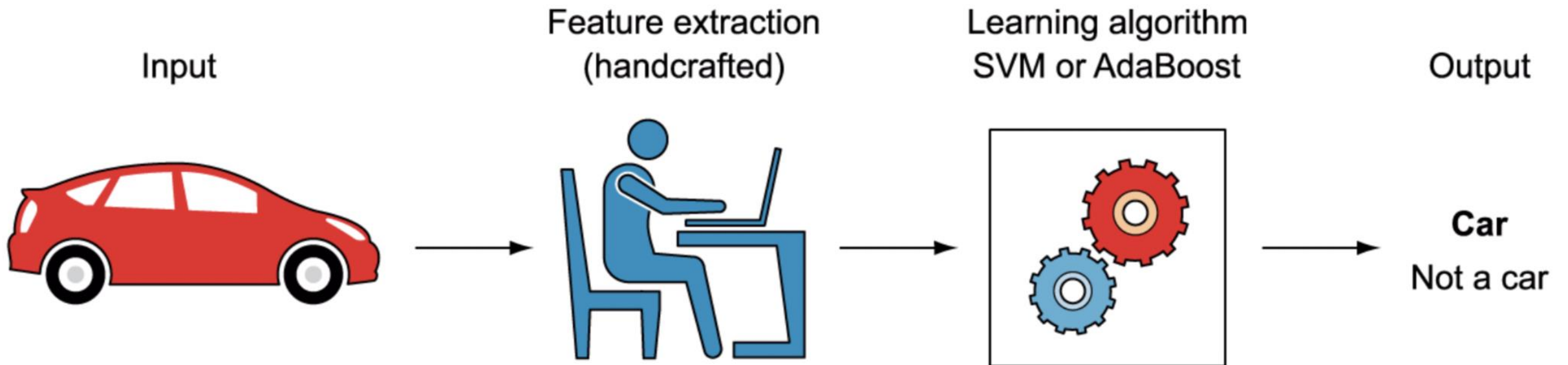
Height Distribution on dog's dataset



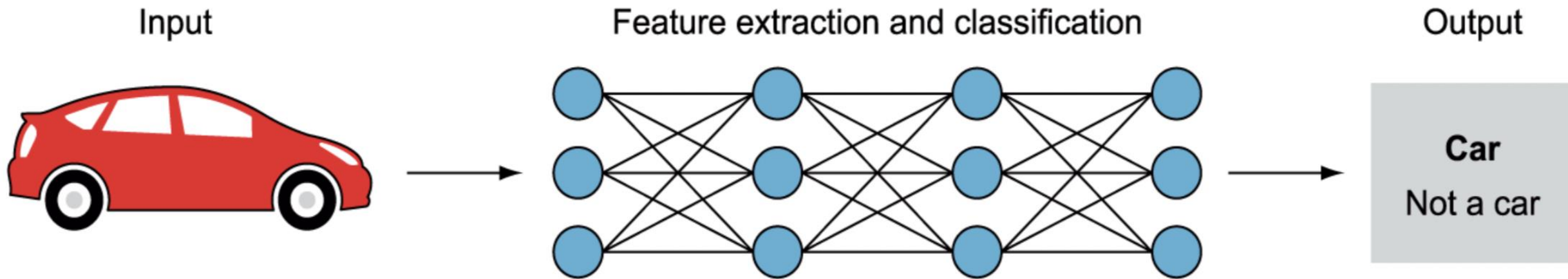
Dog Eye Color distribution on a toy dog dataset



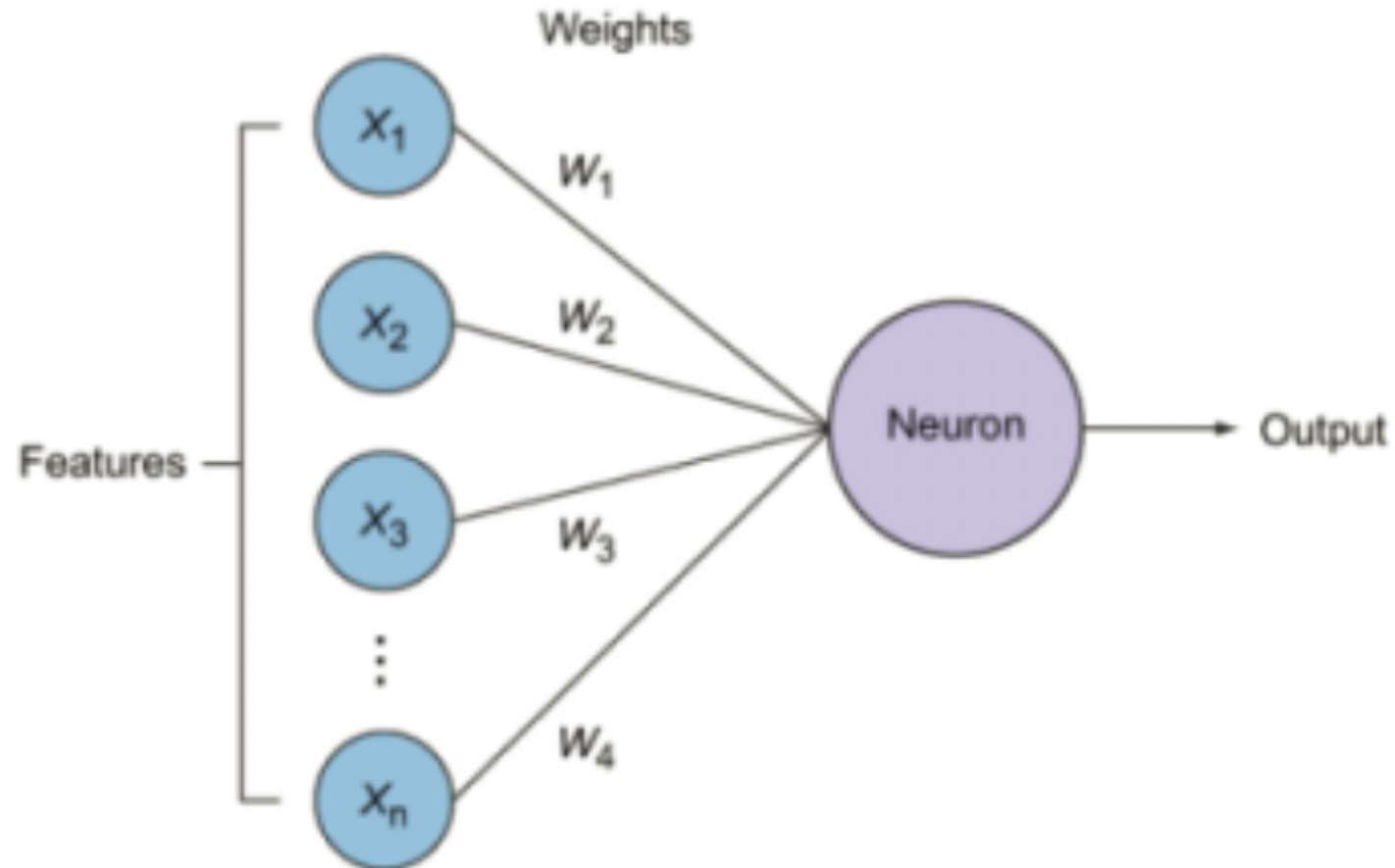
Traditional ML approach for feature extraction



DL approach for feature extraction

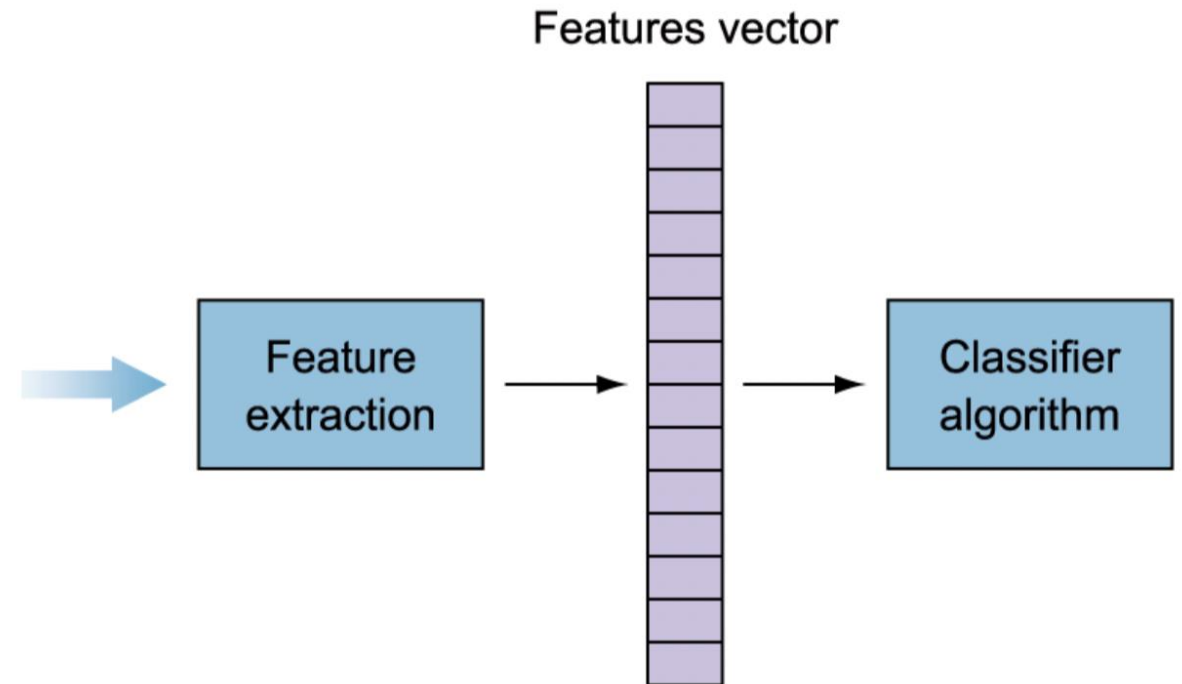
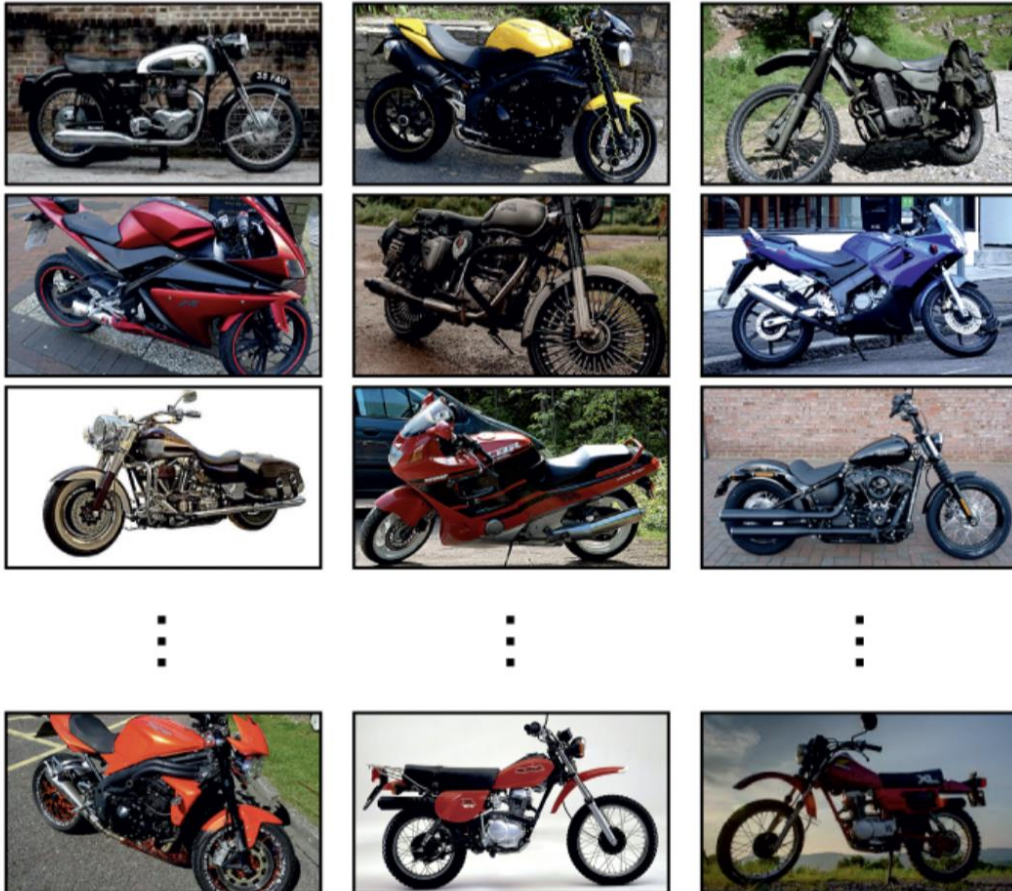


Neural Networks for feature extraction

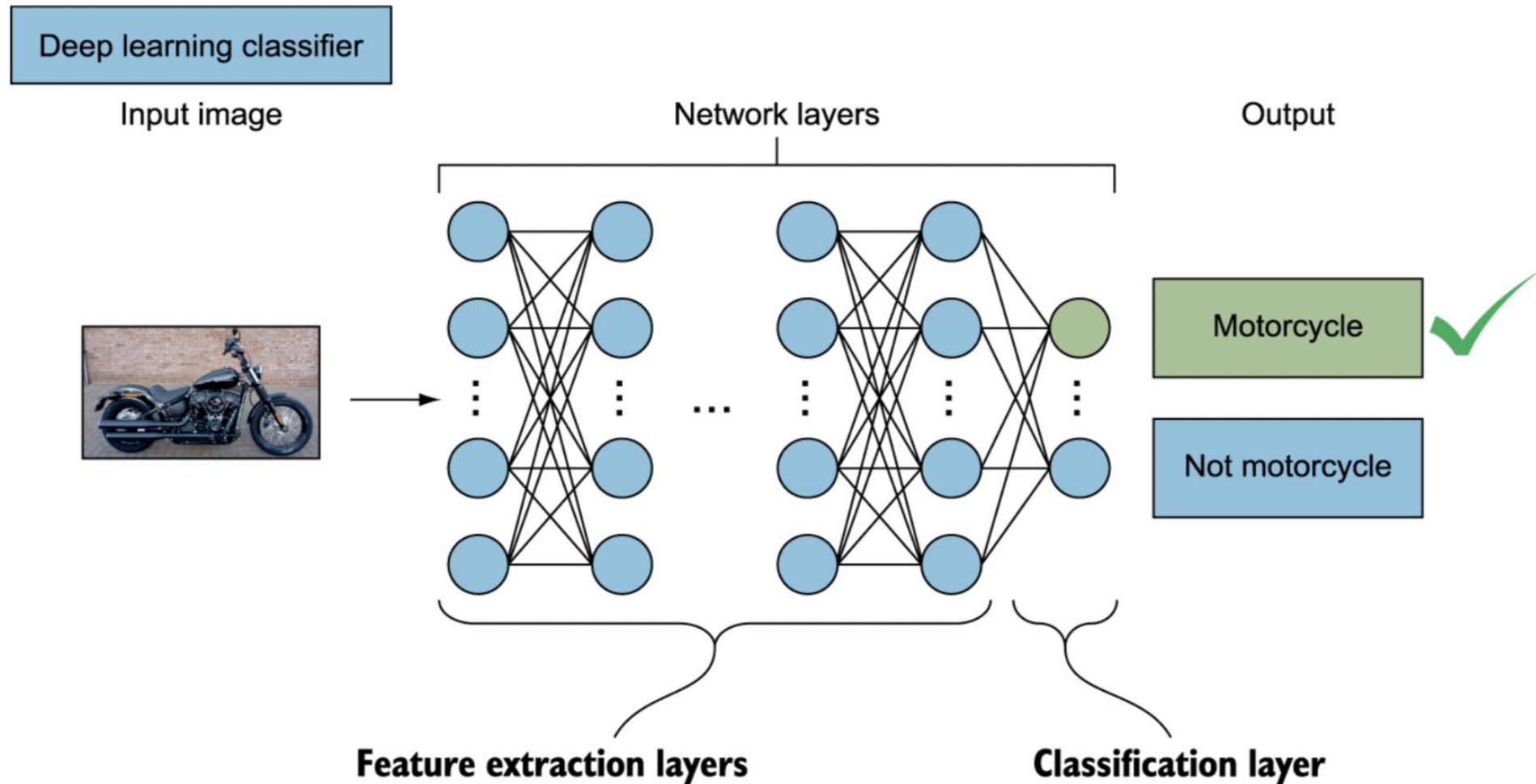


Extracting and Consolidating Features in a Vector

Images dataset of 10,000 images



DL model to extract image features



Reference

- Book – Deep Learning for Vision Systems by Mohamad Elgendy, Manning Publication, November 2020